

BanglaMix: A Multi-Dialect Bengali Speech Dataset and Benchmark for Code-Switched ASR

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Abstract—Automatic speech recognition (ASR) remains difficult for speech that combines low-resource regional dialects with natural code-switching. We introduce BanglaMix, a large-scale silver-standard benchmark containing 41,499 clips (131.2 hours) from 15 Bangladeshi regional dialects, including naturally occurring Bengali–English code-switching. An automated pipeline performs speech segmentation, Whisper-based transcription, script-aware code-switch detection, and LLM-assisted quality filtering. This work compares Whisper, Meta MMS, and wav2vec2-XLS-R systems under zero-shot and fine-tuned conditions. With conventional surface normalization that retains all decoder repetitions, fine-tuned wav2vec2 obtains the lowest silver-reference WER (0.861), while joint-data Whisper obtains 0.964 WER and improves substantially over zero-shot Whisper-small (1.142; $p < 0.001$, two-sided paired bootstrap). As an explicit system-level post-processing condition, repetition collapse reduces joint-data Whisper to 0.801 WER and 0.490 CER; this score is statistically indistinguishable from the Bengali-only Whisper ablation ($p = 0.911$). We therefore make no claim that joint training improves general recognition accuracy: across silver-reference WER, human-reference WER, and LLM-judged quality, the joint-data system is never clearly best, and the three data ablations are statistically indistinguishable under all three. The claim we do make for joint training is narrower and confined to code-switching behaviour: relative to Bengali-only training it raises CS-F1 from 0.011 to 0.192 and English-token survival from 0.01 to 0.13, while retaining the broad-coverage Bengali data that the code-switch-only ablation discards. On a 100 clip two-annotator validation subset, MMS-1B gives the lowest WER against both human references (0.757 and 0.697), demonstrating why silver and human-reference results must be interpreted separately. These findings position BanglaMix as a transparent, challenging benchmark for studying dialect robustness, decoder repetition, and English-token preservation rather than as a solved ASR task.¹

Index Terms—Automatic speech recognition, Bengali ASR, code-switching, dialectal speech, low-resource speech recognition, Whisper, and a large language model judge.

I. INTRODUCTION

Speech is the most natural way for people to interact with technology. However, automatic speech recognition (ASR) systems do not work equally well for all languages and speaking styles. Two important reasons for poor ASR performance are dialect variation and code-switching (CS), where speakers mix words from another language within the same sentence. These are common in everyday Bangladeshi speech.

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¹Dataset: https://huggingface.co/datasets/niloycste68/Bangali_local_dialect_ASR_HF_Dataset — gated access pending the institutional determinations recorded in the release register; code: <https://github.com/niloycste/ASR-Bengali-Local-Dialect->.

Bangla is one of the most widely spoken languages in the world with approximately 280–310 million speakers globally and about 170 million speakers in Bangladesh (where it is spoken by over 98% of the population), and has a rich set of regional dialects. In Bangladesh alone, it is estimated that there are more than 40–50 dialects, including major varieties such as Chittagonian, Sylheti, Noakhali, Rangpuri, and Barisal, which differ significantly in pronunciation, vocabulary, and expression. Informal speakers may mix English words with dialectal Bengali. For example, a speaker may say, ‘*chatgaiya bhashay phone unboxing kortachi, honestly speaking camera performance onek bhalo*’ meaning ‘unboxing a phone in the Chittagong dialect; honestly speaking, the camera performance is very good.’

Despite this, most existing Bengali speech datasets focus on only one problem at a time. For example, read-speech datasets like OpenSLR-37 [1] mainly cover standard Bengali. The Ben-10 corpus [2] includes ten regional dialects but does not include code-switching. On the other hand, the MUCS shared task [3] provides Bengali–English code-switching data, but only in standard Dhaka Bengali. We are not aware of another released resource that jointly targets broad regional variation and naturally occurring Bengali–English switching at this scale. This gap between real-world speech and available resources makes it difficult to build robust ASR systems.

Recognizing dialectal code-switched speech is difficult for several reasons. First, regional dialects can be quite different from standard Bengali in pronunciation, sound patterns, rhythm, and vocabulary, so models trained on standard Bengali do not perform well. Second, code-switching requires the system to handle smooth switching between Bengali and English within the same sentence, often mixing languages that standard monolingual models are not designed for. Third, there is very little training data for dialectal and code-switched speech, and manually transcribing data across 15 dialects is very expensive. Finally, many dialect words and borrowed English terms do not have fixed spellings, so even correct transcriptions may be marked wrong by strict evaluation metrics. These challenges together motivate the design of our dataset, pipeline, and evaluation approach. They lead us to build an automated annotation pipeline without manual transcription, use script-aware code-switch detection, adopt a shared vocabulary sequence-to-sequence model, and focus on character-level and code-switch-aware evaluation metrics.

This work fills this gap; Fig. 1 provides an end-to-end overview of the dataset construction, models, and evaluation. The main contributions of this paper are as follows:

- 1) One of the first large-scale resources designed specifically for the intersection of regional Bengali dialects and

Bengali–English code-switching: 41,499 silver-transcribed clips (131.2 hours) across 15 dialects.

- 2) A scalable automated construction pipeline using voice-activity segmentation, Whisper-based silver transcription, script-based code-switch detection, and multilingual LLM review of borderline cases, complemented by a two-annotator validation subset.
- 3) A diagnostic benchmark and ablation study comparing three ASR model families under zero-shot and fine-tuned settings, including controlled training subsets (dialect-only, code-switching-only, and combined) to measure the contribution of each data type.
- 4) A detailed evaluation and error analysis across dialects and code-switching conditions, including two-sided significance tests, bootstrap uncertainty for code-switching metrics, and an LLM-ensemble meaning-preservation diagnostic.

BanglaMix is positioned primarily as a dataset-construction and auditability contribution. The model experiments are diagnostic stress tests of the released resource, not an exhaustive state-of-the-art leaderboard. Their purpose is to measure failure modes under a common protocol and to separate model behavior from silver-reference noise, which is independently estimated in Section III-E.

The rest of the paper is organized as follows: Section II reviews related work on Bengali, dialectal, and code-switched ASR. Section III describes the BanglaMix dataset and its construction process. Section IV presents the ASR models and fine-tuning strategy, while Section V details the experimental setup. Section VI reports the main results, Section VII provides detailed error analysis, ablation studies, and case studies, Section VIII discusses limitations and reproducibility, and Section IX concludes the paper with future directions.

II. RELATED WORK

This section reviews prior studies on Bengali automatic speech recognition (ASR), covering standard and dialectal ASR, code-switched speech recognition, large scale pretrained speech models, automated speech corpus construction, and large language model (LLM)-based data quality assessment. While recent advances in multilingual and low-resource ASR have improved speech recognition performance across many languages, existing Bengali ASR research has largely focused on standard Bengali speech. As a result, the combined challenges of regional dialect variation and Bengali–English code-switching remain underexplored.

A. Bengali and Dialectal ASR

Early Bengali ASR systems were mainly based on Gaussian mixture models and deep neural network hidden Markov models with manually designed pronunciation lexicons [4], [5]. These systems mostly focused on standard Bengali and had limited ability to handle unseen words. Later, Bengali ASR research moved toward end to end approaches, including connectionist temporal classification (CTC) [6], deep recurrent networks [7], Deep Speech [8], and attention based encoder decoder models such as Listen, Attend and Spell [9]. Toolkits

such as ESPnet [10] further supported this progress and helped reduce error rates for standard Bengali ASR [11]. More recently, the Transformer [12] and masked pre training [13] have enabled powerful self supervised and weakly supervised speech models, including wav2vec.2.0 [14], HuBERT [15], WavLM [16], XLS-R [17], [18], Whisper [19], and MMS [20]. Together with multilingual datasets such as Common Voice [21], these models have improved transfer learning for resource limited Bengali speech recognition. However, dialectal Bengali has received focused attention only recently. The Ben-10 corpus [2], which contains more than 78 hours of speech from 394 speakers across ten regional dialects, shows that even strong foundation models such as Whisper large v3 perform poorly on dialectal speech that differs from the training data, both before and after fine tuning. Biswas *et al.* [22] propose a WavLM based denoising and adaptation framework for dialectal Bengali, including Chittagonian and Sylheti, under simulated noise, and report large relative reductions in WER compared with wav2vec.2.0 and Whisper at low signal to noise ratios. Ridoy *et al.* [23] compare Whisper and Wav2Vec-BERT on standard Bangla and find that the self supervised encoder is both more accurate and more computationally efficient in resource limited settings. However, these studies do not address code switching, which motivates the need for a benchmark that brings dialect variation and code switching together in a single resource.

B. Code-Switching ASR

Code switching, where speakers alternate between two languages within a conversation or even within a single sentence, is common in multilingual communities. It is especially frequent in informal Bangladeshi speech, where English words are often inserted into Bengali sentences. This creates several challenges for ASR systems. First, the model must recognize words from two languages and, in the case of Bengali and English, two different writing systems. Second, code switched speech is still rare in standard ASR training datasets. Third, language switching often occurs together with accent, pronunciation, and acoustic variation, which makes both speech recognition and language modeling more difficult [24], [25]. Progress in code switched ASR has largely been supported by dedicated datasets, such as the Mandarin English SEAME corpus [26]. However, recent studies show that even large multilingual models do not handle code switching reliably unless they are trained with explicit code switched data [27]. For Indian languages, the MUCS 2021 challenge [3] introduced Bengali English and Hindi English code switching benchmarks, but the Bengali data is limited to standard Dhaka Bengali and does not include regional dialects. One useful feature of Bengali English code switching is that the two languages use different Unicode ranges. This allows word level language identification to be performed almost directly from the script, without requiring a separate language identification model. From a modeling perspective, sequence to sequence models such as Whisper are better suited to this setting because their shared subword vocabulary [28] can represent both scripts and generate mixed script text in a single decoding

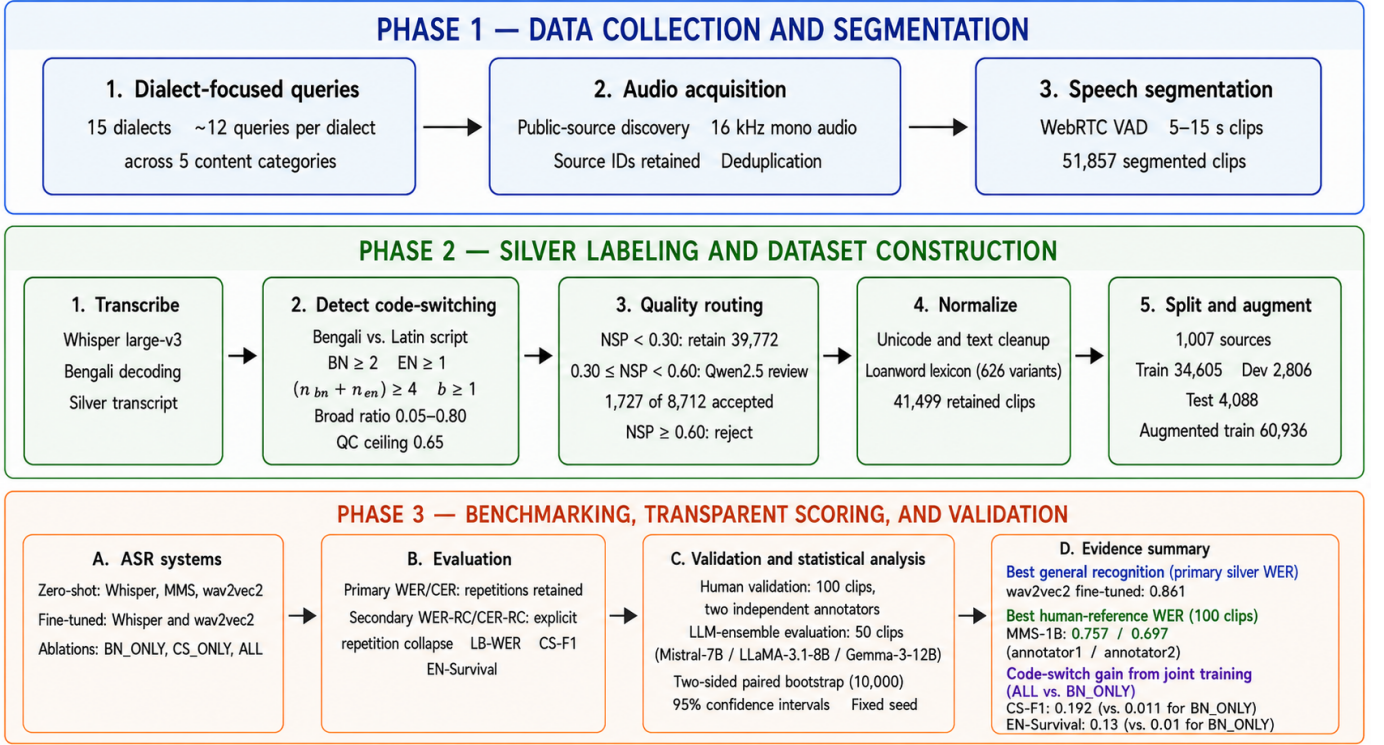


Fig. 1. Auditable overview of dataset construction and evaluation. Clips with no-speech probability below 0.30 are retained, clips in $[0.30, 0.60)$ receive LLM review, and clips at or above 0.60 are rejected. Evaluation separates conventional surface-normalized scores, which retain repetitions, from explicitly labeled repetition-collapsed system scores.

process. In contrast, monolingual CTC models such as MMS are usually restricted to one output script and therefore cannot properly represent code switched speech. This work evaluates both modeling paradigms to measure this gap in dialectal Bengali ASR.

C. Automated Corpus Construction for Low-Resource ASR

Since manual transcription is difficult to scale for many low-resource language varieties, recent speech corpora increasingly rely on automated data construction pipelines. For example, GigaSpeech 2 [29] builds a large multi-domain corpus for Thai, Indonesian, and Vietnamese using automated web crawling, Whisper-based transcription, and iterative refinement. This shows that weakly supervised labels can support large-scale corpus creation without relying entirely on human transcribers. Other studies have also used synthetic speech and spectrogram augmentation to increase the amount of training data for low-resource ASR [30]. BanglaMix follows this automated approach, but focuses on a previously underexplored setting: multi-dialect Bengali speech with English code switching. It also adds an LLM-based judge to filter borderline or uncertain transcripts.

D. LLM-as-a-Judge for Data Quality

Large language models (LLMs) are increasingly used as automatic evaluators in natural language processing, including pairwise response judging [31], [32] and multi-dimensional

quality scoring [33]. These models are especially useful when human annotation is expensive and reference labels are limited or noisy, as in low-resource dialectal ASR. However, their reliability depends on careful prompt design and the capability of the judging model. In this work, we use LLM judges at two stages. During dataset construction, a strong multilingual model, Qwen2.5 [34], evaluates borderline silver-standard transcripts across several quality dimensions and decides whether they should be retained. During evaluation, an ensemble of three open models, Mistral [35], LLaMA [36], and Gemma [37], provides an additional reference-free assessment that complements word-level metrics, as discussed in Section VII. To the best of our knowledge, using LLM judges for both quality control and evaluation in code-switched low-resource ASR has not been previously explored.

Qwen2.5 was selected because its open weights permit fixed, local inference: transcripts need not be sent to a commercial API, and the same model and prompt can be rerun by other researchers. This choice is operational rather than a claim that Qwen2.5 is superior to commercial judges or manual review. Moreover, the construction-stage judge receives text only; it assesses transcript plausibility and cannot verify acoustic alignment. The threshold cascade and the independent human sample are therefore essential safeguards rather than treating the LLM decision as ground truth.

Recent multilingual systems provide additional context for the experimental scope. SeamlessM4T supports Bengali as an input speech language and is a relevant future cross-

TABLE I

SUMMARY OF REPRESENTATIVE BENGALI AND CODE-SWITCHING ASR RESOURCES AND MODELS. “CS” DENOTES CODE-SWITCHING SUPPORT, AND “–” INDICATES NOT APPLICABLE. BANGLAMIX IS, TO OUR KNOWLEDGE, THE FIRST RESOURCE TO COMBINE MULTI-DIALECT BENGALI WITH ENGLISH CODE-SWITCHING.

Ref.	Method / Model	Objective	Variety	CS	Hours	Key limitation
Hasnat et al. [4]	GMM-HMM	Isolated-word Bengali ASR	Standard BN	No	–	Lexicon-dependent; WER >50%; no OOV handling
Swarna et al. [5]	Survey (HMM/DNN)	Bengali phoneme recognition survey	Standard BN	No	–	Survey; standard Bengali, no dialect or CS
Mandal et al. [11]	End-to-end (CTC/attention)	Lexicon-free Bengali ASR	Standard BN	No	–	Standard Bengali only; no dialect or CS
Kjartansson et al. [1]	OpenSLR-37 corpus	Read-speech benchmark	Standard BN	No	100	Read speech; no dialect, no code-switching
MUCS 2021 [3]	CS challenge corpus	Bengali–English CS-ASR	Standard BN	Yes	16	Standard Dhaka Bengali; no dialect
Ben-10 [2]	Whisper fine-tuning	Multi-dialect Bengali ASR	Dialectal BN	No	78	No code-switching
XLS-R [17]	wav2vec2 self-supervised	Multilingual transfer	Multilingual	No	–	Standard Bengali via transfer; no dialect/CS
Whisper [19]	Weakly-supervised seq2seq	Multilingual ASR	Multilingual	No	–	~70% WER on unseen Bengali dialects
MMS [20]	CTC with adapters	1000+ language ASR	Multilingual	No	–	Monolingual output; cannot emit code-switching
Ridoy et al. [23]	Whisper vs. Wav2Vec-BERT	Low-resource Bangla model comparison	Standard BN	No	90–100	Standard Bangla only; no dialect or CS
Biswas et al. [22]	WavLM denoising + adaptation	Noise-robust dialectal Bengali ASR	Dialectal BN	No	~20	Dialectal but no code-switching; noise-focused
GigaSpeech 2 [29]	Automated crawl + transcribe + refine	Large-scale low-resource corpus	Th/Id/Vi	No	~30k	Not Bengali; no dialect or code-switching

architecture baseline [38]. Canary-1B is a strong multilingual ASR model, but its released language set is limited to English, German, French, and Spanish and therefore does not provide a Bengali decoding condition [39]. The present suite is consequently representative rather than exhaustive.

E. Comparison with Existing ASR Resources

Table I summarizes representative Bengali and code-switching ASR resources based on the dimensions most relevant to this study. The comparison highlights two clear gaps in the existing literature. First, standard Bengali ASR corpora and models [1], [4], [5], [11] mainly focus on standard speech and do not address either regional dialect variation or code switching. Second, existing specialized resources cover only one of these challenges at a time. The Ben-10 corpus [2] includes multiple Bengali dialects but does not contain code-switched speech, while the MUCS task [3] provides Bengali English code-switching data only for standard Bengali. Although large multilingual models such as XLS-R, Whisper, and MMS [17], [19], [20] improve transfer to Bengali, their performance often drops on dialectal speech. In addition, monolingual CTC models cannot properly generate mixed script code-switched output. BanglaMix addresses this gap by focusing on the intersection of these two underexplored problems: multi-dialect Bengali speech with natural English code switching. To the best of our knowledge, it is one of the first resources designed specifically for this setting.

III. THE BANGLAMIX DATASET CONSTRUCTION

The BanglaMix dataset is designed to study real Bengali speech with regional dialects and natural English code switching. Its main goal is to capture how people actually speak in everyday life. The dataset focuses on three main points: real dialect speech, natural use of English words inside Bengali sentences, and automatic dataset creation without manual transcription. The audio is collected from social media using dialect based search queries, so it reflects authentic regional speech rather than only standard Bengali pronunciation. After collection, the audio is automatically segmented, transcribed, and checked for quality. As a result, BanglaMix provides a large and challenging dataset for testing and improving ASR systems in real world speech conditions.

A. Dataset Design Principles

BanglaMix is built on three main principles. First, it focuses on dialectal authenticity, meaning that the audio should contain real regional dialect speech rather than standard Bengali spoken by people from a particular region. This is important because searching with region names, such as “Chittagong vlog”, often returns videos where speakers from that region use standard Bengali for a general audience. In contrast, searching with dialect names is more likely to find content where speakers intentionally use their local dialect. Second, BanglaMix focuses on natural code-switching, where English words appear naturally in informal speech rather than in scripted examples. This is important because scripted code-switching may not reflect real pronunciation, word choice, or

TABLE II
CONTENT ANGLES USED TO DESIGN DIALECT-NAME SEARCH QUERIES,
AND WHY EACH ELICITS AUTHENTIC DIALECTAL CODE-SWITCHING.

Content type	Reason for dialectal code-switching
Technology/phone review	Forces English technical terms within dialect speech
Street interview/reaction	Spontaneous public responses; a fake accent cannot be sustained
Gaming / live stream	Live commentary with frequent English game terms
Campus/career	Academic and job English mixed into regional speech
Vlog / content creator	Relaxed informal speech with frequent CS

speaking patterns. Third, BanglaMix emphasizes scalability. The dataset construction pipeline is fully automated so that the corpus can be expanded without paid manual transcribers, which is especially important for low-resource languages with limited annotation budgets.

B. Dialect-Aware Search Query Design

Search queries play a central role in collecting authentic dialectal code-switched speech because they determine which creators and speech styles are retrieved. As discussed earlier, region-name queries often return standard Bengali content, so BanglaMix instead uses dialect-name queries combined with content genres where spontaneous code-switching is more likely to occur. For each dialect, around twelve queries are designed across five content categories, as summarized in Table II. The motivation is that street interviews and live reaction videos usually contain spontaneous speech, where speakers are less likely to maintain an artificial accent. Technology and campus-related content often include English technical terms, while vlog content captures relaxed, informal, and naturally mixed speech. To improve data quality, videos uploaded before 2021 and videos with very low viewership are excluded.

C. Construction Pipeline

Algorithm 1 summarizes the end-to-end pipeline; each step is justified below.

Step 1: Data Acquisition and Segmentation: Audio is collected from social media platforms, including YouTube, TikTok, and Facebook, using yt-dlp. All audio is downloaded in 16 kHz mono WAV format. A shared de-duplication archive is used to ensure that the same video is not collected more than once. Since social media recordings often contain long silent or non-speech parts, each recording is divided into shorter 5–15 s speech clips using WebRTC voice activity detection. This clip length is suitable for modern ASR models and helps make both transcription and fine-tuning more efficient. This step produces 51,857 clips. No additional manual filtering is applied at this stage; clips that remain uncertain or noisy are handled later by the no-speech threshold and the LLM-based silver-standard filtering stage.

Algorithm 1 BanglaMix: From raw dialectal audio to a curated CS-ASR benchmark

Input: dialect-name queries $\mathcal{Q}$; VAD aggressiveness a ; no-speech thresholds $\tau_0=0.30$, $\tau_{hi}=0.60$; source-stratified split target $\approx 83/7/10$
Output: curated dataset $\mathcal{D} = (\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{dev}}, \mathcal{D}_{\text{test}})$

Step 1: Acquire and segment
1: $\mathcal{R} \leftarrow \text{DOWNLOAD16K}(\mathcal{Q})$ // yt-dlp + ffmpeg; shared de-duplication archive
2: $\mathcal{C} \leftarrow \text{VADSEGMENT}(\mathcal{R}, a)$ // WebRTC VAD; 5–15 s clips $\rightarrow 51,857$

Step 2: Transcribe
3: **for all** clip $c \in \mathcal{C}$ **do**
4: $t_c \leftarrow \text{WHISPERLARGEV3}(c, \text{lang}=\text{bn})$
5: **end for**

Step 3: Detect code-switching
6: **for all** clip $c \in \mathcal{C}$ **do**
7: Compute n_{bn} , n_{en} , b , and $\rho = n_{\text{en}}/(n_{\text{bn}} + n_{\text{en}})$ from t_c using Bengali and Latin scripts
8: $R_c \leftarrow (n_{\text{bn}} + n_{\text{en}}) \geq 4 \wedge n_{\text{bn}} \geq 2 \wedge n_{\text{en}} \geq 1 \wedge b \geq 1 \wedge 0.05 \leq \rho \leq 0.80$
9: $c.\text{CS} \leftarrow 1[R_c]$
10: **end for**

Step 4: Silver-standard annotation
11: Define $\text{NSP}(c)$ as the Whisper large-v3 no-speech probability for clip c
12: $\mathcal{A} \leftarrow \{c : \text{NSP}(c) < \tau_0\}$ // auto-accept $\rightarrow 39,772$
13: $\mathcal{M} \leftarrow \{c : \tau_0 \leq \text{NSP}(c) < \tau_{hi}\}$ // borderline band $\rightarrow 8,712$
14: **for all** $c \in \mathcal{M}$ **do**
15: $d \leftarrow \text{QWEN2.5JUDGE}(t_c)$ // 4 yes/no criteria; ACCEPT/REJECT/UNCERTAIN
16: **if** $d = \text{ACCEPT}$ **then**
17: $\mathcal{A} \leftarrow \mathcal{A} \cup \{c\}$ // +1,727
18: **else**
19: d do not add c to $\mathcal{A}$ // REJECT/UNCERTAIN clips conservatively discarded
20: **end if**
21: **end for**

Step 5: Normalize transcripts
22: **Final silver-standard clips:** $|\mathcal{A}| = 41,499$
23: **for all** $c \in \mathcal{A}$ **do**
24: $t_c \leftarrow \text{NORMALIZE}(t_c)$ // NFC; lexicons; repeated-character collapse; URL/emoji cleanup
25: **end for**

Step 6: Leakage-safe source-stratified split
26: $(\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{dev}}, \mathcal{D}_{\text{test}}) \leftarrow \text{SOURCESTRATSPLIT}(\mathcal{A})$ // source-level; dialect-stratified; no cross-split augmentation
27: **Split sizes:** 34,605/2,806/4,088 // $\approx 83/7/10$

Step 7: Augment training split only
28: $\mathcal{G}_{\text{train}} \leftarrow \text{AUGMENT}(\mathcal{D}_{\text{train}})$ // speed $\times 0.9 / \times 1.1$; noise (20 dB); pitch ± 2 semitones
29: $\mathcal{D}_{\text{train}}^{\text{aug}} \leftarrow \mathcal{D}_{\text{train}} \cup \mathcal{G}_{\text{train}}$
30: **Training clips after augmentation:** 60,936
31: **return** $\mathcal{D}$ // released dataset; $\mathcal{D}_{\text{train}}^{\text{aug}}$ used for model training

Step 2: Transcription: Each clip is transcribed using Whisper large-v3 [19], with the language setting forced to Bengali. Whisper large-v3 is used as the automatic labeler because it is one of the strongest openly available multilingual ASR models and provides the best available automatic substitute for manual transcription in this difficult setting.

Step 3: Code-Switch Detection: Code-switching is detected using Unicode script information. Bengali words use the Bengali Unicode range (U+0980–U+09FF), while English words usually appear in Latin script. Since these scripts are separate, word-level language identification can be performed almost directly from the script, without using a separate language identification model.

Let n_{bn} be the number of Bengali-script words, n_{en} be the number of Latin-script words, b be the number of language boundaries between adjacent words of different scripts, and

$\rho = n_{\text{en}} / (n_{\text{en}} + n_{\text{bn}})$ be the English-word ratio. A clip is labeled as code-switched if it satisfies the following conditions:

$$\begin{aligned} (n_{\text{bn}} + n_{\text{en}}) &\geq 4, & n_{\text{bn}} &\geq 2, & n_{\text{en}} &\geq 1, \\ b &\geq 1, & 0.05 &\leq \rho \leq 0.80. \end{aligned} \quad (1)$$

A stricter high-confidence subset is also created by requiring $n_{\text{en}} \geq 2$ and $0.10 \leq \rho \leq 0.70$. During automatic annotation, an additional conservative QC check excludes CS candidates with $\rho > 0.65$ from automatic acceptance. Thus, 0.80 is the detector’s broad labeling boundary, whereas 0.65 is the release-stage automatic-acceptance ceiling. The broad detector rule is applied to model outputs when computing CS-F1.

Step 4: Silver-Standard Annotation with an LLM Judge: Since large-scale manual transcription of dialectal code-switched speech is not practical, BanglaMix uses a two-stage silver-standard annotation strategy. We define $\text{NSP}(c)$ as the Whisper large-v3 no-speech probability for clip c , used as a confidence proxy for speech presence and transcript reliability. Clips with $\text{NSP}(c) < \tau_{\text{o}} = 0.30$ are retained automatically as high-confidence speech, following prior evidence that high-confidence Whisper transcripts can provide useful training labels [29]. Clips with $\text{NSP}(c) \geq \tau_{\text{hi}} = 0.60$ are rejected as low-confidence or non-speech clips. The remaining clips, with $0.30 \leq \text{NSP}(c) < 0.60$, are treated as borderline cases.

For the borderline set, Qwen2.5 [34] evaluates each transcript using the same local Ollama-based prompt. Open weights make the decision rule reproducible while keeping transcripts off third-party APIs. The prompt asks four yes/no questions: whether the transcript is valid Bengali rather than hallucinated text, whether Bengali–English mixing is natural, whether it resembles dialectal or informal spoken Bengali, and whether it resembles speech rather than noise, repeated text, or song lyrics. The implementation returns ACCEPT when three or four criteria pass, UNCERTAIN for a two–two split, and REJECT when zero or one criterion passes. The parser derives this outcome from the four criterion values rather than trusting the model’s self-reported overall label. Only ACCEPT clips enter the release.

The stored pre-judge and final manifests make the cascade auditable. Retaining only $\text{NSP} < 0.30$ would yield 39,772 clips, whereas accepting every borderline candidate would yield 48,484. Qwen2.5 retains 1,727 of the 8,712 borderline clips (19.8%) and excludes 6,985 (80.2%), giving the 41,499-clip final set. These counts quantify the judge’s practical effect, but not its accuracy. The judge has no access to audio and cannot establish acoustic correctness; no superiority over a commercial LLM or manual review is claimed. The two-annotator study in Section III-E instead provides an independent estimate of residual silver-label noise.

Step 5: Transcript Normalization: The transcripts are cleaned using several normalization steps. These include Unicode NFC normalization, removal of zero-width characters, URLs, hashtags, mentions, and emojis, digit normalization, repeated character reduction, and punctuation and whitespace cleanup. For evaluation only, different spellings of borrowed English words are also mapped to one standard form using a curated lexicon with 271 entries and 650 spelling variants. This step is important because dialectal Bengali does not have

a fixed writing system for many loanwords. As a result, the same spoken word can be written in several acceptable ways. Without this mapping, WER could become higher because of valid spelling differences rather than real recognition errors.

The pipeline also uses three supporting lexicons to handle dialectal and code-switched speech more accurately. First, a list of about 5,700 known ASR hallucination and noise tokens is used to detect corrupted transcripts during normalization. Second, a list of about 18,900 valid dialectal Bengali word forms is protected from normalization into standard Bengali, so genuine dialect words are preserved. Third, a list of about 3,700 common English words helps identify code-switched tokens. Together, these lexicons make the normalization process suitable for the two main targets of this dataset: regional dialect speech and Bengali–English code-switching.

Step 6: Leakage-Safe Source-Stratified Split: After silver-standard filtering and normalization, the 41,499 accepted clips are split directly at the source-video level; no additional DROPLowQUALITY stage is applied after the LLM filtering step. Keeping all clips from the same original video in the same split prevents speaker or recording-channel leakage and avoids artificially improving test performance. The split is also stratified by dialect, so each dialect appears in the training, validation, and test sets. The final split contains 34,605 training clips, 2,806 development clips, and 4,088 test clips, corresponding to approximately 83/7/10%.

Step 7: Training Augmentation: Audio augmentation is applied only to the training data to improve robustness against different acoustic conditions. The augmentation methods include speed perturbation ($\times 0.9$ and $\times 1.1$), additive noise at 20 dB, and pitch shifting by ± 2 semitones, which complement spectrogram-level methods such as SpecAugment [40]. Augmented clips keep the same source information as their original clips and are included only in the training split. This ensures that augmentation does not leak any source across different splits. After augmentation, the training set contains 60,936 clips.

D. Dataset Statistics

Table III presents the 15 dialects included in BanglaMix, and Fig. 2 shows the distribution of clips across these dialects. The distribution is broad but naturally imbalanced, reflecting the availability of dialectal content on social media. From 51,857 segmented clips, corresponding to approximately 164 hours of raw audio, 41,499 clips pass the silver-standard annotation process. These accepted clips form the released dataset and contain 131.2 hours of speech. The remaining approximately 33 hours are removed by the no-speech threshold or the LLM judge. Table IV reports the train, development, and test splits.

The main strength of the dataset is its dialect coverage, with 15 regional varieties represented over 131.2 hours of audio. Code-switching appears as a natural minority, covering 5.7% of the clips, because English insertion occurs spontaneously in real-world speech rather than at a fixed artificial rate. Therefore, BanglaMix preserves the natural frequency of code-switching instead of oversampling it. In this work, code-switching is treated as a separate evaluation dimension on top

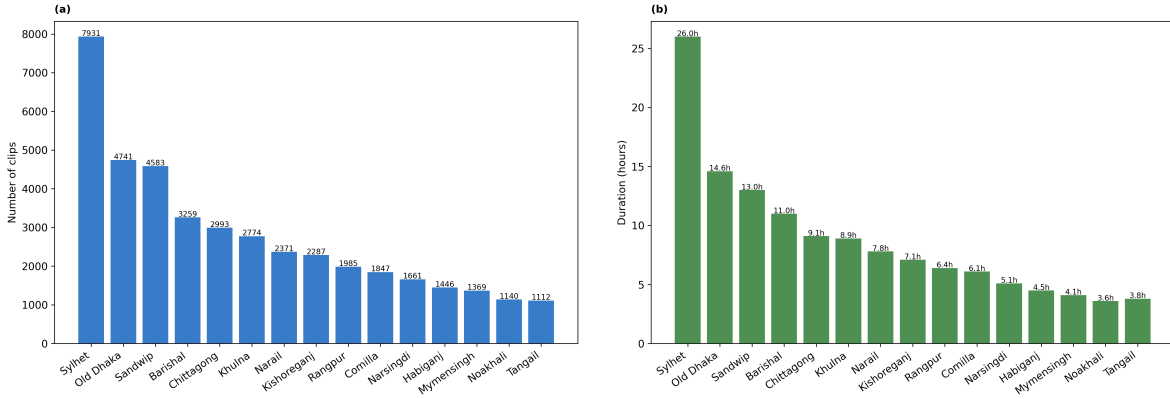


Fig. 2. Dialect distribution across the released dataset: (a) number of clips and (b) total duration in hours.

TABLE III
DIALECT COVERAGE IN THE RELEASED DATASET.

Dialect	Clips	Hours
Sylhet	7,931	26.0
Old Dhaka	4,741	14.6
Sandwip	4,583	13.0
Barishal	3,259	11.0
Chittagong	2,993	9.1
Khulna	2,774	8.9
Narail	2,371	7.8
Kishoreganj	2,287	7.1
Rangpur	1,985	6.4
Comilla	1,847	6.1
Narsingdi	1,661	5.1
Habiganj	1,446	4.5
Mymensingh	1,369	4.1
Noakhali	1,140	3.6
Tangail	1,112	3.8
Total	41,499	131.2

TABLE IV
TRAIN/DEV/TEST SPLITS OF THE RELEASED (REAL, UNAUGMENTED) DATASET. AUDIO AUGMENTATION EXPANDS THE TRAINING SPLIT TO 60,936 CLIPS FOR THE MODEL TRAINING, WHILE DEV AND TEST REMAIN REAL AND UNAUGMENTED. BECAUSE THE SPLIT IS SOURCE-STRATIFIED, THE FINAL PROPORTIONS ARE APPROXIMATELY 83/7/10 RATHER THAN EXACT.

Split	Clips	Hours	CS clips	CS%
Train	34,605	108.9	2,019	5.8
Dev	2,806	9.1	166	5.9
Test	4,088	13.2	176	4.3
Total	41,499	131.2	2,361	5.7

of broad dialect coverage, and CS-specific results are reported on this naturally smaller subset in Section VI.

E. Human Validation of Silver-Standard Labels

As the released transcripts are silver-standard labels, we evaluate them against independent human transcriptions. Two human annotators listened to the same 100 test clips and transcribed the speech independently without access to each other’s annotations. Both annotators completed all 100 clips.

The silver transcript is treated as the hypothesis and each human transcript as a reference. Every score in this section, in Table V, and in Table XVII is produced by the same `primary_normalize` routine that defines the primary condition of the silver-reference benchmark: surface cleaning only, with no pronunciation-variant mapping and no repetition collapse. Human-reference and silver-reference numbers are therefore measured under one identical scoring condition.

Against the first annotator, the silver labels obtain a corpus WER/CER of 0.73/0.44; against the second annotator, they obtain 0.80/0.49. These values estimate silver-label noise but should not be interpreted as model accuracy against a fully adjudicated gold reference.

Across all 100 clips, the two human transcripts have a normalized exact-match rate of 7.0%. Using the first annotator as the reference, the second annotator has WER/CER of 0.60/0.38; using the second annotator as the reference, the first annotator has WER/CER of 0.52/0.34. The directional difference is expected because edit-rate denominators depend on the reference length. The low exact-match rate and moderate character-level disagreement reflect both genuine listening differences and legitimate orthographic variation in dialectal Bengali.

The per-dialect results in Table V show substantial variation. The two annotators also differ in which dialects they find difficult, demonstrating that dialectal spelling and listening uncertainty contribute to the measured label noise. Since most dialects have only five to nine clips, these breakdowns are indicative rather than definitive. We use the silver labels as noisy supervision and for relative benchmarking, while the human transcripts provide an independent validation sample rather than an adjudicated gold set. Table VI gives de-identified excerpts from this sample, illustrating both the retained code-switching spans and a clear silver-label failure.

F. Ethical and Legal Considerations

The source material is publicly accessible social-media content, but public accessibility alone does not establish permission to redistribute audio or eliminate creators’ privacy interests. We therefore retain source identifiers and timestamps, omit speaker names, handles, faces, contact information, and

TABLE V
SILVER-LABEL QUALITY AGAINST TWO INDEPENDENT HUMAN
REFERENCES ON THE 100-CLIP VALIDATION SAMPLE. LOWER IS BETTER;
 n_{A1} AND n_{A2} ARE THE NUMBERS OF COMPLETED TRANSCRIPTS FOR
ANNOTATOR 1 AND ANNOTATOR 2.

Dialect	n_{A1}	WER _{A1}	CER _{A1}	n_{A2}	WER _{A2}	CER _{A2}
Barishal	8	0.73	0.45	8	0.73	0.43
Chittagong	7	0.89	0.53	7	0.76	0.44
Comilla	5	0.94	0.48	5	0.86	0.47
Habiganj	7	1.00	0.66	7	0.89	0.56
Khulna	6	0.68	0.32	6	0.76	0.45
Kishoreganj	8	0.58	0.32	8	0.79	0.42
Mymensingh	9	0.84	0.49	9	0.84	0.43
Narail	6	0.71	0.31	6	0.67	0.29
Narsingdi	6	0.76	0.43	6	0.81	0.44
Noakhali	5	1.09	0.64	5	0.76	0.42
Old Dhaka	8	0.60	0.37	8	0.86	0.70
Rangpur	8	0.75	0.54	8	0.83	0.63
Sandwip	5	0.38	0.35	5	0.85	0.68
Sylhet	6	0.43	0.31	6	0.88	0.72
Tangail	6	0.76	0.47	6	0.75	0.41
Overall	100	0.73	0.44	100	0.80	0.49

demographic labels, and restrict the intended use to non-commercial ASR research. The corpus contains 1,007 source records: 36,967 clips in the general collection and 4,532 in the UK-diaspora collection. These counts describe collection domains, not population strata.

No consented age, gender, socioeconomic, education, or urban–rural metadata are available, and we do not infer sensitive attributes from voices or online profiles. Consequently, BanglaMix cannot support demographic subgroup-fairness claims. Platform visibility, search-query design, and internet access may overrepresent digitally connected, urban, younger, or diaspora creators; these characteristics are plausible selection mechanisms, not measured labels. Dialect coverage therefore reflects online availability rather than population prevalence. Fig. 2 reports the observable dialect imbalance so that users can apply weighting or resampling, while future consent-based collection should record self-reported demographic variables for a proper fairness audit.

The release follows a rights-aware governance procedure. It provides a documented takedown and opt-out channel, prohibits speaker identification, surveillance, and re-identification, and records provenance at the source level.

We state the current clearance status explicitly, because describing a governance procedure is not the same as having completed it. Three determinations are treated as blocking for any public redistribution of audio: institutional confirmation of whether formal ethics review is required, legal or institutional review of the redistribution basis for each source platform and content category, and a recorded source-level allow/remove decision. At the time of writing these three determinations are pending, and they are tracked with named owners and dated evidence in the `ETHICS_AND_DATA_GOVERNANCE.md` register released with the code.

Neither the public accessibility of the source material nor the non-commercial licence we attach to the compilation authorizes redistribution of third-party social-media audio; only those determinations can. Until they are recorded, the

audio release is provisional and the intended distribution is the metadata-only fallback: source pointers, timestamps, clip-level split manifests with checksums, transcripts where permitted, and reproducible processing code, without hosted audio. If review later declines a source, that source is removed under a versioned removal log so mirrors can honour the decision.

These safeguards do not remove all risk: speech is potentially identifying, creators did not provide study-specific consent, and models may perform unevenly across dialects. We therefore report these limitations explicitly, provide per-dialect results, and maintain a removal process throughout the lifetime of the resource.

IV. ASR MODELS AND FINE-TUNING

This work evaluates three ASR model families. These models are selected to represent the two main ASR approaches and different types of pre-training.

Whisper (Sequence-to-Sequence): Whisper [19] is an encoder-decoder transformer model trained on 680k hours of weakly supervised multilingual audio. It is important for this work because its subword vocabulary includes both Bengali and Latin-script tokens. This allows Whisper to generate mixed-script, code-switched output in a single decoding step, making it well suited for Bengali English code-switching. Although convolution-augmented models such as Conformer [41] are also strong for ASR, Whisper is especially useful here because of its shared multilingual vocabulary. For fine-tuning, we use Whisper-small, which has 244M parameters. This model is practical to train on modest hardware while still benefiting from large-scale pre-training. During inference, Bengali transcription is forced for all Whisper models.

Meta MMS (CTC, Multilingual): MMS [20] is a CTC-based multilingual ASR model that supports more than 1,000 languages through language-specific adapters. We include it as a strong multilingual zero-shot baseline. However, its Bengali output vocabulary is monolingual, so it cannot generate English-script words. This makes MMS useful for comparison, because it helps show how much code-switching affects models that are not designed to produce mixed-script output.

wav2vec2-XLS-R (CTC, Self-Supervised): wav2vec2-XLS-R-300M [17] is a self-supervised multilingual speech model pre-trained on 436k hours of audio. We use a Bengali fine-tuned checkpoint as a zero-shot baseline. We also fine-tune the model on BanglaMix so that it can be compared with Whisper using the same training data. This model represents the CTC-based ASR approach in our experiments.

Together, these systems cover multilingual encoder–decoder and CTC paradigms, monolingual versus mixed-script output vocabularies, and zero-shot versus in-domain adaptation. They do not constitute an exhaustive leaderboard. SeamlessM4T is Bengali-capable and would be a useful future comparison, whereas the released Canary-1B vocabulary does not include Bengali [38], [39]. We therefore restrict conclusions to the evaluated model set.

Fine-tuning Setup: Whisper is fine-tuned using the Hugging Face Transformers [42] and Datasets [43] libraries with the Adam optimizer [44]. We use a learning rate of 10^{-5} ,

TABLE VI

DE-IDENTIFIED QUALITATIVE EXAMPLES FROM THE EXISTING 100-CLIP, TWO-ANNOTATOR VALIDATION SAMPLE. EXCERPTS ARE SHORTENED AND LIGHTLY ROMANIZED FOR COMPACT PRESENTATION; THEY ARE ILLUSTRATIONS RATHER THAN NEW ANNOTATIONS.

ID	Dialect	Silver excerpt	Independent-reference excerpt	What it illustrates
E1	Rangpur	“assalamu alaikum to everyone guys, this is your boy ... welcome to my channel!”	“assalamu alaikum to everyone guys, this is your boy ... welcome to my channel!”	Long mixed-language span is retained, with minor Bengali spelling variation.
E2	Comilla	“I think Mainamati museum-e ... What’s inside? Forest?”	“I think Mainamati museum ... What’s inside? Forest?”	Natural intra-utterance switching and an English phrase embedded in Bengali.
E3	Barishal	“eta change hoye jabe; the day after tomorrow ... the day before”	“eta change hoiya jabe; the day after tomorrow ... the day before”	English temporal phrases are preserved despite dialectal spelling variation.
E4	Narsingdi	“emijmen prok ek grem holade prok”	“this is an amusement park, a green holiday park”	Clear silver-label failure, demonstrating why the corpus is not a gold standard.

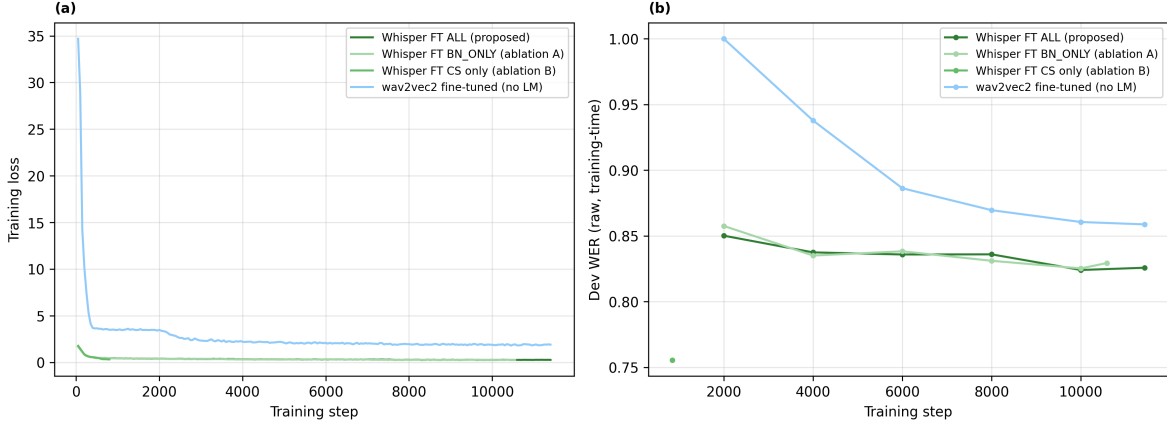


Fig. 3. (a) Training loss and (b) development WER during fine-tuning. Development WER follows the training-time evaluation pipeline and is not directly comparable to the separately reported test WER-RC condition.

an effective batch size of 16, three training epochs, and 500 warm-up steps. The best checkpoint is selected based on development WER. For the CTC model, we build a character vocabulary from the training transcripts, covering Bengali, Latin characters, and common punctuation. The CTC output head is reinitialized, and the encoder is fine-tuned for three epochs with a learning rate of 10^{-4} . All fine-tuning and benchmark inference, including both zero-shot and fine-tuned systems, are run on a single NVIDIA A100-SXM4-40GB GPU. The one-time silver-standard transcription of all clips using Whisper large-v3 is performed on a CPU workstation with an Intel Core i9-14900 processor, 24 cores, and 32 GB RAM.

The CTC model is trained by minimizing the negative log-likelihood over all possible alignments π that collapse to the target sequence y :

$$\mathcal{L}_{\text{CTC}} = -\log \sum_{\pi \in \mathcal{B}^{-1}(y)} \prod_t p(\pi_t | x) \quad (2)$$

where $\mathcal{B}$ is the CTC collapsing function. In contrast, Whisper is trained using the standard token-level cross-entropy loss of its decoder. The main hyperparameters are reported in Table VII. Fig. 3 shows that all Whisper variants converge within about 2k steps. The CS_ONLY ablation runs for fewer

TABLE VII
FINE-TUNING HYPERPARAMETERS FOR THE TWO TRAINED MODEL FAMILIES.

Hyperparameter	Whisper-small	wav2vec2-XLS-R
Base parameters	244M	300M
Architecture	seq2seq	CTC
Learning rate	10^{-5}	10^{-4}
Effective batch size	16	16
Epochs	3	3
Warm-up steps	500	500
Precision	bf16/fp16	bf16/fp16
Model selection	best dev WER	best dev WER

steps because it uses only the approximately 4.5k code-switched clips.

Training Subset Ablation: To understand the effect of different training data types, we train three Whisper variants: BN_ONLY, which uses only dialectal Bengali clips; CS_ONLY, which uses only code-switched clips; and ALL, which uses both types of data. All three models use the same base model, training procedure, and test set. Therefore, differences in performance can be attributed to the type of training data used.

V. EXPERIMENTAL SETUP

This section defines the evaluation metrics, scoring conditions, baseline systems, and statistical tests used for dialectal and code-switched Bengali speech.

A. Evaluation Metrics

We use Word Error Rate (WER) and Character Error Rate (CER) as the main evaluation metrics. CER is especially important in this work because dialectal Bengali does not have a standardized spelling system. As a result, WER may penalize valid transcriptions that use different spellings for the same spoken word, while CER gives a better indication of how closely the model output matches the speech at the character level.

To characterize code-switched speech, we report WER separately for Bengali-script and English-script reference words (WER-BN and WER-EN). Both are computed on the code-switched test clips only, and never on the full corpus, so that they share a scope with the language-balanced WER (LB-WER) defined on the same clips as the unweighted mean

$$\text{LB-WER} = \frac{1}{2}(\text{WER-BN} + \text{WER-EN}). \quad (3)$$

Keeping all three quantities on one subset is what makes the identity above hold in the reported tables; computing the components corpus-wide and the mean on the CS subset would combine incompatible scopes. LB-WER gives the two scripts equal influence despite unequal token counts; it does not test a joint word-and-language label.

Code-Switch Detection F1 (CS-F1) measures whether the hypothesis contains both Bengali and English according to the script-based detector. Unlike the metrics above, CS-F1 is a *clip-level binary classification* score evaluated over the entire test set: all 4,088 clips are classified, of which 176 are code-switched positives and 3,912 are negatives. Precision therefore depends on false positives raised on Bengali-only clips, which is the intended behaviour — a system that emits spurious Latin tokens everywhere should be penalized. CS-F1 is thus not restricted to the 176-clip subset, even though its recall term and the confidence intervals we report are governed by that small positive count.

We distinguish two scoring conditions. The *primary* condition applies conventional surface normalization (Unicode normalization, case handling, punctuation removal, and whitespace cleanup) while retaining every repeated character and word, so decoder insertion errors remain in WER and CER. The *WER-RC/CER-RC* condition additionally collapses excessive character and consecutive-word repetitions and is reported as explicit system post-processing. All models use the same procedure within each condition.

Formally, WER is calculated using substitutions, deletions, and insertions:

$$\text{WER} = \frac{S + D + I}{N} \quad (4)$$

where S , D , and I represent substitution, deletion, and insertion errors, and N is the total number of words. WER can exceed 1.0 when insertion errors are large; in this work,

that happens on a small number of hallucination-heavy zero-shot outputs, so values above 1.0 are expected and do not indicate a scoring bug.

CER is calculated in the same way but at the character level:

$$\text{CER} = \frac{S_c + D_c + I_c}{N_c} \quad (5)$$

CER complements WER because dialectal spellings may differ even when their character sequences remain similar; it is not treated as a replacement for human-reference evaluation.

For code-switch detection, we treat it as a yes/no problem: whether code-switching appears in the output or not. We use Precision, Recall, and F1 score:

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad \text{CS-F1} = \frac{2PR}{P + R} \quad (6)$$

Finally, English-word survival (EN-Surv) measures how many English words in the reference are correctly preserved in the model output for code-switched speech:

$$\text{EN-Surv} = \frac{\sum_{(r,h) \in \mathcal{P}} |\{w \in r : \text{English}(w)\} \cap h|}{\sum_{(r,h) \in \mathcal{P}} |\{w \in r : \text{English}(w)\}|} \quad (7)$$

where $\mathcal{P}$ is the set of code-switched (reference, hypothesis) pairs.

In the discussion below, EN-Surv is our primary code-switching metric because it directly measures English-token retention. WER-CS, WER-BN, WER-EN and LB-WER are reported as supporting diagnostics computed on the smaller 176-clip code-switched test subset; CS-F1 is a detection score over the whole 4,088-clip test set, and only its 176 positives limit the precision of its recall term.

B. Baselines

We compare our method with four zero-shot baseline models. These include Whisper-small (used as a base model), Whisper large-v3 (the largest evaluated Whisper checkpoint), MMS-1B, and wav2vec2-XLS-R-BN, which is trained on standard Bengali speech.

C. Significance Testing

To quantify uncertainty, we use two-sided paired bootstrap tests [45] with 10,000 clip-level resamples. For each resample, corpus WER is recomputed from the summed edit and reference-word counts. The two-sided p -value is twice the smaller bootstrap tail probability around zero, with a plus-one Monte Carlo correction; percentile intervals provide 95% confidence intervals. Tests use a fixed seed and are reported separately for primary WER and WER-RC.

VI. RESULTS

Table VIII and Fig. 4 compare zero-shot and task-adapted systems under both primary and repetition-collapsed scoring.

The zero-shot models show limited robustness: Whisper large-v3 obtains 0.930 primary WER and Whisper-small obtains 1.142. Fine-tuning yields complementary gains. Joint-data Whisper reduces the latter to 0.964 (15.6% relative), while

TABLE VIII

SILVER-REFERENCE RESULTS ON THE BANGLAMIX TEST SET. WER, WER-RC AND CER ARE CORPUS-WIDE OVER ALL 4,088 TEST CLIPS; WER RETAINS ALL REPETITIONS, WHILE WER-RC APPLIES EXPLICIT REPETITION-COLLAPSE POST-PROCESSING. WER-BN, WER-EN, LB-WER AND WER-CS ARE ALL SCOPE TO THE SAME 176 CODE-SWITCHED TEST CLIPS, SO LB-WER IS EXACTLY THE EQUAL-WEIGHT MEAN OF THE WER-BN AND WER-EN COLUMNS IN THE SAME ROW. CS-F1 IS A CLIP-LEVEL DETECTION SCORE OVER ALL 4,088 CLIPS (176 POSITIVES). BOLD NUMERIC VALUES ARE COLUMN MINIMA (OR THE MAXIMUM FOR CS-F1); THE BOLD MODEL NAME IDENTIFIES THE JOINT-DATA WHISPER SYSTEM.

Model	WER	WER-RC	CER	WER-BN	WER-EN	LB-WER	WER-CS	CS-F1
<i>Zero-shot baselines</i>								
Whisper-small	1.142	1.066	1.031	1.000	1.463	1.232	1.133	0.000
Whisper large-v3	0.930	0.910	0.757	0.818	1.106	0.962	0.818	0.144
MMS-1B	0.989	0.990	0.571	1.000	1.131	1.066	0.940	0.041
wav2vec2-XLS-R-BN	0.963	0.963	0.622	0.964	1.039	1.001	0.935	0.044
<i>CTC fine-tuned on BanglaMix</i>								
wav2vec2	0.861	0.861	0.493	0.942	0.996	0.969	0.877	0.135
<i>Whisper fine-tuned on BanglaMix (ablation)</i>								
BN_ONLY (A)	0.923	0.800	0.575	1.047	0.994	1.021	0.933	0.011
CS_ONLY (B)	1.065	0.883	0.679	1.105	0.737	0.921	0.963	0.276
ALL (joint data)	0.964	0.801	0.597	1.106	0.902	1.004	0.959	0.192

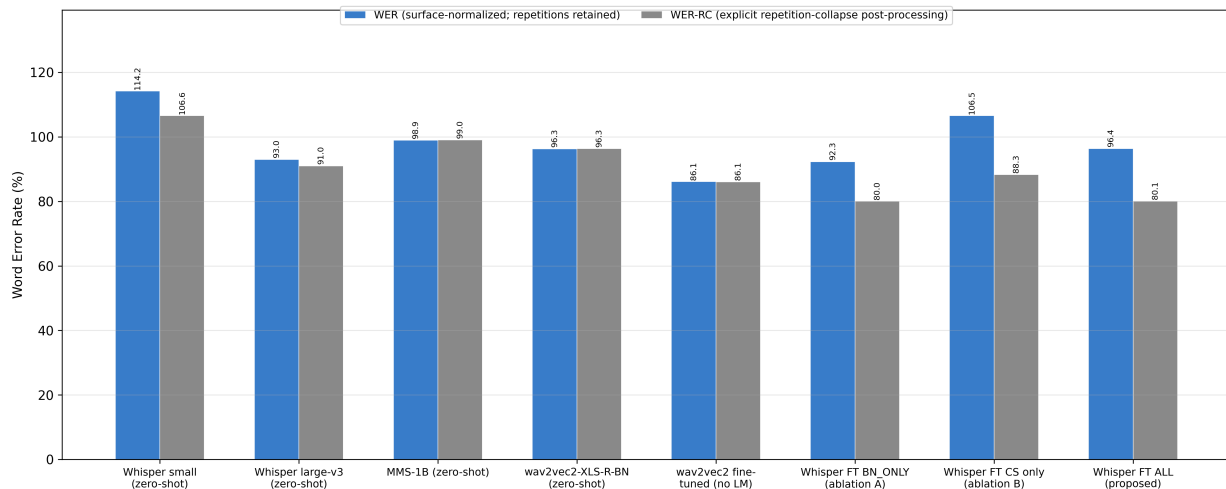


Fig. 4. Surface-normalized WER (repetitions retained) and WER-RC after explicit repetition-collapse post-processing. The large Whisper-specific gap shows why the two conditions must be reported separately.

fine-tuned wav2vec2 gives the lowest primary WER, 0.861. Thus domain adaptation helps, but architecture and decoding behavior remain important.

The Whisper ablations expose a trade-off. BN_ONLY has lower primary WER than ALL (0.923 versus 0.964) but almost no code-switch detection (0.011 CS-F1). CS_ONLY has the strongest English and switching indicators (0.737 WER-EN and 0.921 LB-WER on CS clips; 0.276 CS-F1) but the weakest fine-tuned primary WER (1.065). ALL occupies the middle ground. Its primary WER/CER are 0.964/0.597, while explicit repetition collapse gives 0.801/0.490 WER-RC/CER-RC. The WER-RC score differs from all zero-shot systems and fine-tuned wav2vec2 ($p = 0.0002$), but not from

BN_ONLY ($p = 0.911$); its clearer advantage over BN_ONLY is switching behavior.

A. Per-Dialect Results

Table IX reports the repetition-collapsed Whisper ALL scores, while Fig. 5 and Table X retain repetitions and therefore show primary per-dialect WER. Both conditions reveal substantial dialect-to-dialect variability.

Table X compares representative models under primary scoring. Fine-tuned wav2vec2 has the lowest overall WER and leads on six dialects; Whisper ALL leads on six, and large-v3 leads on three. This heterogeneous ranking motivates per-dialect reporting rather than a single universal claim.

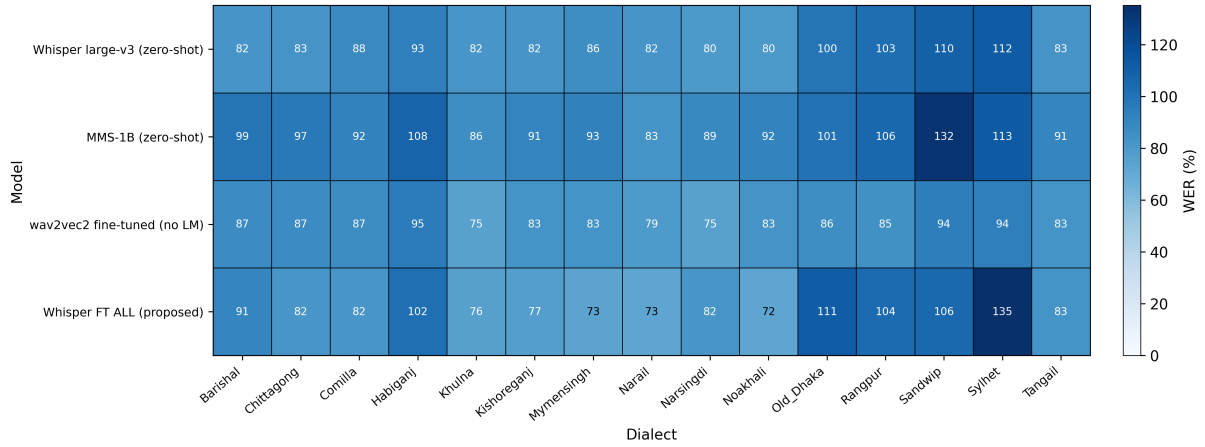


Fig. 5. Primary surface-normalized per-dialect WER for four representative systems; repetitions are retained and darker cells indicate higher WER. Complete eight-model results remain available in the released metrics.

TABLE IX
PER-DIALECT WER-RC/CER-RC FOR WHISPER ALL AFTER EXPLICIT
REPETITION-COLLAPSE POST-PROCESSING.

Dialect	Clips	WER-RC	CER-RC
Sylhet	1,111	0.93	0.65
Old Dhaka	360	0.78	0.53
Sandwip	176	0.92	0.67
Barishal	282	0.87	0.54
Chittagong	198	0.79	0.46
Khulna	120	0.66	0.39
Kishoreganj	743	0.75	0.36
Comilla	157	0.78	0.42
Narail	464	0.70	0.41
Rangpur	165	0.79	0.51
Narsingdi	60	0.72	0.39
Mymensingh	56	0.70	0.35
Habiganj	75	0.90	0.52
Tangail	51	0.74	0.39
Noakhali	70	0.70	0.36
Overall	4,088	0.80	0.49

TABLE X
PRIMARY PER-DIALECT WER FOR REPRESENTATIVE MODELS;
REPETITIONS ARE RETAINED. BOLD = BEST PER DIALECT.

Dialect	large-v3	wav2vec2	Whisper ALL
Sylhet	1.124	0.936	1.352
Old Dhaka	0.997	0.861	1.114
Sandwip	1.096	0.941	1.057
Barishal	0.821	0.872	0.907
Chittagong	0.829	0.872	0.819
Khulna	0.817	0.753	0.756
Kishoreganj	0.822	0.832	0.770
Comilla	0.883	0.866	0.817
Narail	0.822	0.792	0.728
Rangpur	1.028	0.850	1.041
Narsingdi	0.799	0.750	0.817
Mymensingh	0.863	0.829	0.734
Habiganj	0.935	0.951	1.021
Tangail	0.827	0.831	0.831
Noakhali	0.803	0.825	0.724
Overall	0.930	0.861	0.964

TABLE XI
PRIMARY-SCORE ERROR RATES PER REFERENCE WORD AND
ENGLISH-WORD SURVIVAL (EN-Surv); REPETITIONS ARE RETAINED.

Model	S	D	I	EN-Surv
Whisper large-v3 (zs)	0.35	0.45	0.14	0.65
Whisper-small (zs)	0.32	0.67	0.14	0.12
MMS-1B	0.66	0.13	0.19	0.00
wav2vec2 (zs)	0.52	0.36	0.08	0.00
wav2vec2 (FT)	0.53	0.23	0.10	0.00
Whisper BN_ONLY	0.53	0.13	0.26	0.01
Whisper CS_ONLY	0.62	0.10	0.34	0.37
Whisper ALL	0.54	0.12	0.31	0.13

VII. ANALYSIS AND DISCUSSION

This section analyzes model behavior beyond overall error rates. It examines error patterns, code-switching behavior, and qualitative failure cases across different models and dialects. It also evaluates how well models preserve code-switching structure, compares model outputs with human references, and reports statistical significance. The goal is to understand where the improvements come from and which parts of dialectal and code-switched ASR remain difficult.

A. Error Types and English-Word Survival

Table XI separates WER into three error types: substitutions (S), deletions (D), and insertions (I). It also reports English-word survival (EN-Surv), which measures the proportion of English words in the reference that are correctly preserved in the model output for code-switched clips. Fig. 6 shows the substitution, deletion, and insertion breakdown visually.

A clear pattern is observed. Monolingual CTC models and the Bengali-only ablation preserve almost no exact English tokens. CS_ONLY reaches the highest EN-Surv among fine-tuned models (0.37), while ALL reaches 0.13. Large-v3 preserves the most English tokens overall (0.65) and obtains 0.930 primary WER, whereas ALL obtains 0.964 primary WER but 0.801 WER-RC. Thus joint-data fine-tuning improves switching relative to BN_ONLY, while decoder repetition remains a separate weakness.

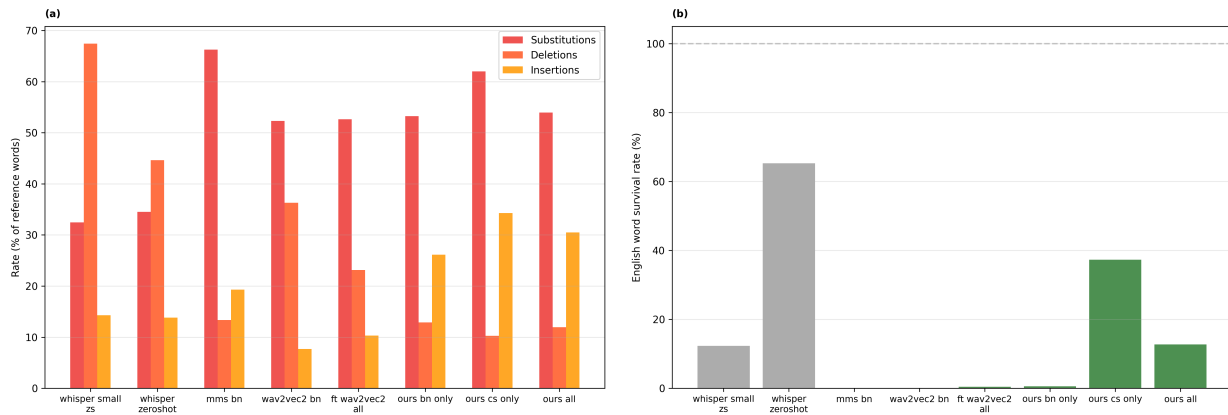


Fig. 6. Error analysis by model: (a) substitution, deletion, and insertion rates per reference word; and (b) English-word survival rate on code-switched clips.

B. Code-Switching Error Analysis

We first clarify the scope of the code-switching results. The per-clip CS error rates — WER-CS, WER-BN, WER-EN and LB-WER — are computed on only 176 code-switched test clips. CS-F1 is scoped differently: every one of the 4,088 test clips is classified as code-switched or not, so its false positives come from the 3,912 Bengali-only clips even though only 176 positives are available to estimate recall. Therefore, these results should be treated as indicative rather than highly precise. However, the main code-switching finding does not depend only on this small subset. The strongest pattern is qualitative and very clear: monolingual and Bengali-only models almost completely drop English words. This English-deletion effect is visible across the English-token counts in Table XI, so it is not just a small difference caused by limited sample size.

Because only 176 test clips are labeled as code-switched, whole-clip WER-CS and the recall side of CS-F1 are treated as descriptive supporting measures rather than definitive rankings. English-word survival provides a directly interpretable complementary measure, but it likewise requires confirmation on a larger adjudicated subset.

Bootstrap intervals from the existing predictions quantify this uncertainty. For Whisper ALL, WER-CS is 0.959 (95% CI [0.868, 1.063]), CS-F1 is 0.192 ([0.120, 0.264]), and English-word survival is 0.127 ([0.064, 0.201]). CS_ONLY has the higher CS-F1, 0.276 ([0.220, 0.328]). The paired CS-F1 difference for ALL minus BN_ONLY is +0.181 ([0.111, 0.253]), whereas ALL minus CS_ONLY is −0.083 ([−0.153, −0.015]). Thus, existing evidence supports the narrower conclusion that mixed-data training improves switching over Bengali-only training, not that it surpasses code-switch-only training on CS-F1.

Table XII gives a descriptive comparison of primary WER on code-switched and Bengali-only clips, and Fig. 7 shows the same contrast alongside the detection scores. Most gaps are small or negative, so this comparison does not support a causal claim that the CS subset is intrinsically harder; the subsets differ in content and length. The clearer finding is token-specific: English survival is near zero for monolingual and BN_ONLY systems, while CS-supervised Whisper systems recover more mixed-script output.

TABLE XII
PRIMARY WER ON CS VS. BENGALI-ONLY CLIPS. GAP = WER-CS − WER-BN. THE BOLD MODEL NAME IDENTIFIES THE JOINT-DATA SYSTEM.

Model	WER-CS	WER-BN	Gap
Whisper large-v3 (zs)	0.818	0.936	−0.118
Whisper-small (zs)	1.133	1.142	−0.009
MMS-1B	0.94	0.99	−0.05
wav2vec2 (FT)	0.877	0.860	+0.017
Whisper CS_ONLY	0.963	1.071	−0.108
Whisper ALL	0.959	0.964	−0.005

Beyond checking whether English words are preserved, we also evaluate whether models can reproduce the structure of code-switching. Walking through a transcript from left to right, we label every language-bearing token as Bengali or English and count a *switch* at each adjacent pair whose labels differ; the 176 code-switched test clips contain 463 such reference switch points. Preservation is the capped recall $\sum_c \min(\hat{s}_c, s_c) / \sum_c s_c$, where s_c and $\hat{s}_c$ are the reference and hypothesis switch counts of clip c ; the cap prevents a model from scoring highly by over-generating Latin tokens. We report this analysis only for the Whisper sequence-to-sequence models, because they can naturally generate mixed Bengali and Latin script. Monolingual CTC models such as MMS are excluded because they cannot reliably produce Latin-script words, with EN-survival of 0.00 in Table XI. Therefore, any apparent switches in their output are likely character-level artifacts rather than real code-switching. To reduce such artifacts, a Latin token is counted as English only if it contains at least two letters.

Table XIII reports this analysis. Reference code-switched clips contain an average of 2.63 language switches, and every Whisper system produces fewer. Within the fine-tuned family, preservation tracks the amount of code-switch supervision exactly as expected: CS_ONLY preserves 36% of reference switches, ALL 12%, and BN_ONLY almost none (2%). Zero-shot large-v3 is highest overall (46%), consistent with it also having the highest English survival in Table XI. Two cautions apply. Switch density does not test exact boundary placement, so a model can match the count while placing

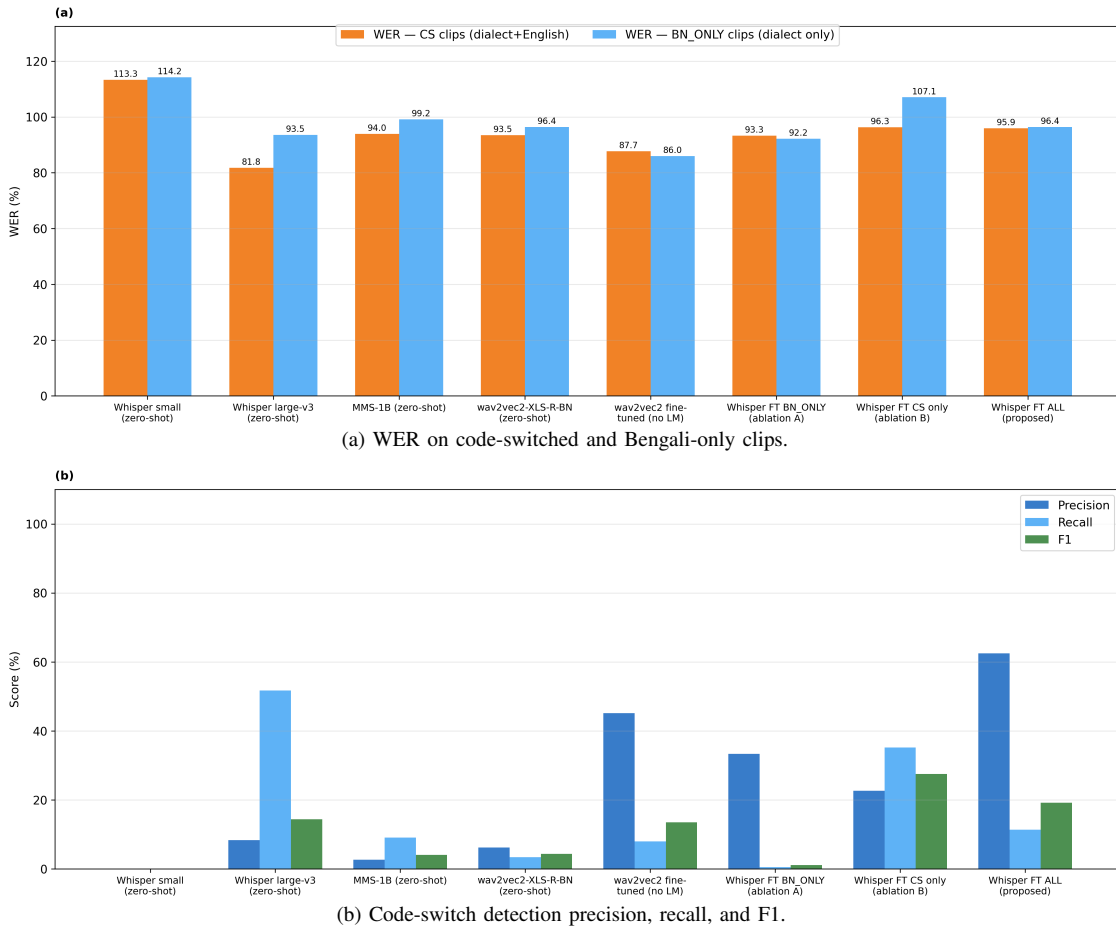


Fig. 7. Code-switching analysis: (a) surface-normalized WER on code-switched and Bengali-only clips; and (b) code-switch detection precision, recall, and F1. Models never trained on CS show near-zero CS-F1.

TABLE XIII
CODE-SWITCH STRUCTURE PRESERVATION ON THE 176 CS TEST CLIPS (WHISPER SEQ2SEQ FAMILY): AVERAGE INTRA-UTTERANCE LANGUAGE SWITCHES PRODUCED, AND THE FRACTION OF THE 463 REFERENCE SWITCH POINTS PRESERVED, $\sum_c \min(\hat{s}_c, s_c) / \sum_c s_c$. REFERENCE CLIPS AVERAGE 2.63 SWITCHES. COMPUTED BY EVALUATION/17_CS_STRUCTURE_PRESERVATION.PY.

Model	Sw/clip	Preserved
Reference	2.63	—
Whisper large-v3 (zs)	1.50	46%
Whisper CS_ONLY	1.35	36%
Whisper ALL	0.31	12%
Whisper BN_ONLY	0.05	2%

switches at the wrong words, and the measure may still be inflated by over-generation that the cap does not fully remove. It is therefore reported as a descriptive upper bound on positional preservation. The gap between CS_ONLY and ALL also shows that joint training buys code-switch detection (Table VIII) at a real cost in reproduced switch structure.

C. Qualitative Error Analysis

Inspection of model outputs shows three repeated error patterns. For readability, the examples below use romanized Bengali and omit Bengali script.

- Repetition hallucination occurs when Whisper generates repeated words on difficult clips. For example, a short reference such as “tai” may be expanded into many repeated tokens. Primary WER counts every insertion; WER-RC reports the behavior of an explicitly post-processed system.
- English deletion occurs when monolingual or Bengali-focused models remove English words from code-switched speech. For example, the reference “ami office-e meeting korlam” may be decoded as “ami ... korlam”, where the English words are missing. This explains the near-zero English-word survival shown in Table XI.
- Spelling variation occurs when fine-tuned models produce phonetically correct words using different spellings, such as “kortasi” and “kortechi” for the same spoken form. These cases are counted as word-level errors, but they do not always indicate real recognition failure. This also explains why CER is much lower than WER.

These patterns explain the main trends in the results. Fine-tuning mainly reduces hallucination and English-word deletion, while many remaining errors come from acceptable spelling differences caused by the lack of standardized dialect spelling.

Table XIV expresses substitutions, deletions, and insertions as shares of all primary word errors. Large-v3 has a 48% dele-

TABLE XIV

ERROR TAXONOMY SHOWING DELETION, INSERTION, AND SUBSTITUTION SHARES, PLUS BENIGN SPELLING VARIANTS, EACH AS A SHARE OF ALL WORD ERRORS ($S + D + I$) RECOVERED FROM THE JIWER ALIGNMENT. A SPELLING VARIANT IS A SUBSTITUTION WHOSE NORMALIZED CHARACTER EDIT DISTANCE $\text{lev}(r, h) / \max(|r|, |h|)$ IS ≤ 0.34 . COMPUTED BY EVALUATION/18_SPELLING_VARIANT_ANALYSIS.PY; SUCH PAIRS ARE REPORTED, NEVER RELABELED AS CORRECT.

Model	Del	Ins	Sub	Spell. var.
Whisper large-v3 (zs)	48%	15%	37%	5%
wav2vec2 (FT)	27%	12%	61%	15%
Whisper ALL	12%	32%	56%	14%

TABLE XV

EXPLICIT SYSTEM POST-PROCESSING ABLATION FOR WHISPER ALL. PRIMARY WER/CER RETAIN REPETITIONS; WER-RC/CER-RC COLLAPSE DEGENERATE RUNS.

Configuration	WER	CER
Primary: repetitions retained	0.964	0.597
System output: repetition collapse	0.801	0.490

tion share. For ALL, the deletion share falls to 12%, but the insertion share rises to 32%, exposing the repetition behavior hidden by WER-RC. Approximately 14–15% of errors in the fine-tuned systems satisfy the near-spelling threshold, against 5% for zero-shot large-v3, so fine-tuning shifts error mass from outright deletion toward orthographic near-misses. These are reported as potential orthographic variants, not automatically relabeled as correct.

D. Ablation Study

Two ablations isolate the training subset and system post-processing. BN_ONLY has stronger primary WER than ALL but almost no code-switch detection; CS_ONLY has stronger English retention but substantially weaker overall WER. Repetition collapse reduces ALL from 0.964 to 0.801 WER and from 0.597 to 0.490 CER without changing model parameters. Because this operation removes genuine insertion errors, the uncollapsed result remains primary and the collapsed result is labeled WER-RC/CER-RC throughout.

E. Case Study

A single Sylheti test clip illustrates decoder repetition: a one-word reference is expanded by zero-shot Whisper-small into roughly forty repeated tokens. Primary scoring correctly counts these as insertions. Repetition-collapse post-processing shortens such degenerate runs, explaining the WER-RC change in Table XV; it does not constitute an improvement to the underlying acoustic model. A second recurring failure is English-token deletion by monolingual and Bengali-only systems.

F. Statistical Significance

The recomputed two-sided paired-bootstrap results separate the two scoring conditions. For primary WER, ALL improves

over zero-shot Whisper-small ($p = 0.0002$) and MMS-1B ($p = 0.0066$), is indistinguishable from the zero-shot wav2vec2 Bengali checkpoint ($p = 0.9411$), and is worse than large-v3 ($p = 0.0012$), fine-tuned wav2vec2 ($p = 0.0002$), and BN_ONLY ($p = 0.0002$). For WER-RC, ALL has a 95% interval of $[0.789, 0.813]$ and differs from every zero-shot baseline, fine-tuned wav2vec2, and CS_ONLY ($p = 0.0002$ in each comparison), but not from BN_ONLY ($p = 0.9109$). These results support a specific claim: repetition-controlled Whisper is strong on the silver benchmark, while raw decoding accuracy is led by fine-tuned wav2vec2. The code-switch-specific bootstrap intervals reported in Section VI are wider, reflecting only 176 positive test clips, and therefore support directional comparisons rather than a universal ranking.

G. LLM-Ensemble Evaluation

WER measures surface-level transcription accuracy, but it does not fully show whether the output keeps the correct meaning or preserves code-switching. Therefore, we use an additional reference-free evaluation with an ensemble of three open LLMs: Mistral-7B [35], LLaMA-3.1-8B [36], and Gemma-3-12B [37]. Each model scores the ASR output from 0 to 5 on three aspects: semantic correctness, code-switch preservation, and dialect naturalness.

All four judged systems are scored on one shared 50-clip sample rather than on an independent draw per system. The sample is built once with a fixed seed, stratified into 25 code-switched and 25 Bengali-only clips and spread across dialects within each stratum, and its SHA-256 digest is released so that a reviewer can verify that every system was judged on the identical clip list. Code-switched clips are deliberately over-represented relative to their 4.3% test-set rate, because an unstratified 50-clip draw would contain roughly two code-switched clips and leave the code-switch dimension uninformative. Sharing the clip list is what makes the per-clip differences in Table XVI paired, so they can be tested directly instead of being read off means computed on different data. We use the same two-sided paired bootstrap as Section VI, with 10,000 resamples and a fixed seed.

All three fine-tuned Whisper systems score above zero-shot large-v3 on this diagnostic, which reverses the ordering that primary WER gives. BN_ONLY has the highest overall score (2.89/5) and leads every individual dimension, ALL follows at 2.81, CS_ONLY at 2.68, and large-v3 is lowest at 2.47. The paired tests show that only the fine-tuned-versus-zero-shot gaps are reliable at this sample size: large-v3 differs from BN_ONLY (-0.418 , $p = 0.0118$) and from ALL (-0.338 , $p = 0.0348$), but not from CS_ONLY (-0.209 , $p = 0.2710$). The three ablations are mutually indistinguishable: BN_ONLY versus ALL is $+0.080$ ($p = 0.1532$), BN_ONLY versus CS_ONLY is $+0.209$ ($p = 0.0602$), and CS_ONLY versus ALL is -0.129 ($p = 0.2018$). The ALL-versus-BN_ONLY result therefore agrees with the WER-RC comparison in Section VI, where the same two systems were also statistically indistinguishable ($p = 0.911$). Every system scores lower on the code-switched half of the sample than on the Bengali-only half, and the gap is largest for zero-shot large-v3 (2.13

TABLE XVI

LLM-ENSEMBLE EVALUATION (MEAN SCORE 0–5; THREE OPEN JUDGES) ON ONE SHARED, CODE-SWITCH-STRATIFIED 50-CLIP SAMPLE (25 CS / 25 BN) JUDGED IDENTICALLY BY ALL FOUR SYSTEMS. “AGREE” IS THE MEAN FRACTION OF JUDGE PAIRS SCORING WITHIN ONE POINT. BOLD = BEST PER COLUMN; THE CODE-SWITCH-PRESERVATION COLUMN IS REFERENCE-FREE AND IS DISCUSSED WITH THE CAVEAT IN THE TEXT.

Model	Sem.	CS-pres.	Dialect	Overall	Agree
Whisper large-v3 (zs)	2.40	2.43	2.57	2.47	0.713
Whisper BN_ONLY (A)	2.91	3.03	2.73	2.89	0.811
Whisper CS_ONLY (B)	2.74	2.73	2.56	2.68	0.789
Whisper ALL (joint)	2.82	2.92	2.68	2.81	0.787

versus 2.81) and smallest for BN_ONLY (2.69 versus 3.08). Pairwise within-one-point agreement among the three judges ranges from 0.71 to 0.81 and is lowest for large-v3.

One result requires explicit caution and bounds how this table should be read. BN_ONLY obtains the highest code-switch-preservation score (3.03) even though it has the lowest code-switch detection of any Whisper system (0.011 CS-F1, Table VIII) and preserves essentially no English tokens (0.01 EN-Surv, Table XI). The two measurements do not conflict; they ask different questions. The judges are reference-free and text-only: they see the hypothesis but neither the reference transcript nor the audio, so a fluent Bengali-script output that silently deleted every English word still reads as well-formed. This dimension therefore scores how plausible the language mixing in the output looks, not whether the output kept the English that was actually spoken. Fidelity to the spoken code-switching is measured by the reference-based EN-Surv, CS-F1, and switch-preservation results, and those remain the basis for every code-switching claim in this paper. More generally, LLM judges may share model biases, the sample contains 50 clips, and this diagnostic is not a human mean-opinion score or preference study; it complements but does not override WER or human-reference evaluation.

H. Interpreting High Error Rates

Primary WER remains high for every system, ranging from 0.861 to 1.142. Values above 1.0 arise when insertions exceed the number of reference words. This reflects difficult dialectal speech, non-standard spelling, decoder failures, and noise in Whisper-derived silver references. CER and WER-RC illuminate specific aspects of that difficulty, but neither removes the need for primary WER and human-reference evaluation. BanglaMix is therefore positioned as a stress-test benchmark with transparent scoring conditions.

I. Human-Reference Evaluation

Since the test labels are silver-standard, the main benchmark may favor systems whose output resembles Whisper. We therefore evaluate every model against both human transcripts in the 100-clip validation sample, under the same primary scoring condition used for Table VIII. The ranking changes substantially. MMS-1B has the lowest WER and CER against both annotators (0.76/0.39 and 0.70/0.36), followed by fine-tuned wav2vec2; the whole Whisper family, including the proposed joint-data system, falls behind both. Against the

TABLE XVII

MODEL ERRORS ON THE 100-CLIP VALIDATION SAMPLE USING THE TWO INDEPENDENT HUMAN REFERENCES. BOTH ANNOTATORS COMPLETED ALL 100 REFERENCES. SCORING USES THE SAME PRIMARY CONDITION AS TABLE VIII (SURFACE NORMALIZATION ONLY; NO PRONUNCIATION-VARIANT MAPPING AND NO REPETITION COLLAPSE), SO THE TWO TABLES ARE DIRECTLY COMPARABLE. LOWER VALUES ARE BETTER.

Model	WER _{A1}	CER _{A1}	WER _{A2}	CER _{A2}
MMS-1B (zero-shot)	0.76	0.39	0.70	0.36
wav2vec2 (fine-tuned)	0.85	0.47	0.82	0.47
wav2vec2-XLS-R-BN (zero-shot)	0.87	0.54	0.86	0.57
Whisper large-v3 (zero-shot)	0.92	0.74	0.86	0.71
Whisper BN_ONLY	0.92	0.56	0.86	0.52
Whisper ALL (joint data)	0.93	0.54	0.86	0.50
Whisper CS_ONLY	0.93	0.56	0.86	0.51
Whisper-small (zero-shot)	1.11	1.00	1.03	0.96

first annotator the three Whisper ablations are effectively tied (0.92–0.93), so the CS/BN/ALL ordering seen on the silver benchmark does not survive a human reference. This ranking change is important evidence of reference-dependent bias and is reported as a central result rather than a secondary caveat.

Fine-tuning still improves the Whisper family relative to zero-shot Whisper-small under both human references, and MMS-1B’s result motivates a larger adjudicated evaluation across model families. Because the current sample contains only 100 clips and two independent, non-adjudicated transcripts, the human ranking is informative but not definitive.

VIII. LIMITATIONS

Several limitations bound the claims of this study. First, the transcripts are *silver-standard*: they are generated by Whisper large-v3 and textually filtered by Qwen2.5, which cannot inspect the audio. The two-annotator validation on 100 clips estimates silver-label WER/CER at 0.73/0.44 and 0.80/0.49 against the two independent references, but it is neither large nor adjudicated. Because Whisper generated the silver references, rankings may also favor Whisper-like outputs. The Qwen cascade is reproducible and its retention rate is reported, but its accept/reject accuracy has not been independently human-labeled. The archived construction snapshot stores overall Qwen votes but not all four criterion values; the released script now records those values and derives the decision deterministically. A regenerated release should therefore be versioned rather than assumed to be byte-identical to the archived snapshot.

Second, code-switching is a natural minority: 2,361 of 41,499 clips (5.7%) overall and only 176 clips in the test set. The reported bootstrap intervals expose the resulting uncertainty, and the CS findings should not be generalized beyond Bengali–English switching in these social-media domains. Third, the baseline suite is representative rather than exhaustive. Larger fine-tuned models and Bengali-capable SeamlessM4T may change absolute rankings; Canary-1B is not a Bengali baseline. No cross-corpus evaluation on Ben-10, Common Voice, or another external Bengali test set was run, so this study makes no claim of out-of-domain generalization.

Finally, dialect and source domain are the only available audit dimensions. The absence of consented demographic

metadata prevents age-, gender-, socioeconomic-, and urban-rural fairness analyses. Search and platform selection may overrepresent more visible, digitally connected speakers, and the dialect distribution is imbalanced. The informal vlog, comedy, review, and diaspora domains also differ from broadcast news, healthcare, and education. The LLM-ensemble semantic scores add a meaning-preservation diagnostic, but they remain model-based and cannot substitute for human judgments; being reference-free and text-only, they also cannot detect content that the model silently omitted, which is why code-switching fidelity is claimed only from the reference-based metrics. These limitations motivate consent-based demographic auditing, targeted CS collection, external-corpus testing, and a larger human gold set when resources permit.

IX. CONCLUSION

BanglaMix contributes a large silver-standard benchmark at the intersection of 15 Bengali dialects and Bengali-English code-switching. The revised evaluation separates recognition from post-processing: fine-tuned wav2vec2 gives the best primary silver-reference WER (0.861), while joint-data Whisper gives 0.964 primary WER and 0.801 WER-RC after explicit repetition collapse.

We state the basis for joint training explicitly, because all three of our evaluations agree on what it is not. Under silver-reference WER-RC joint training ties with the Bengali-only ablation ($p = 0.911$); under human references it falls behind MMS-1B and fine-tuned wav2vec2, with the three ablations tied at 0.92–0.93; and under the LLM judge the Bengali-only ablation is nominally highest, with no ablation pair separable (joint versus Bengali-only, $p = 0.153$). Joint training is therefore not a general accuracy improvement, and we do not present it as one.

Its benefit is confined to code-switch behaviour, where it raises CS-F1 from 0.011 to 0.192 and English-token survival from 0.01 to 0.13 over Bengali-only training. That comparison is against the silver reference like the WER results, and it is not a maximum: the code-switch-only ablation reaches a higher EN-Surv (0.37) at the cost of the worst fine-tuned WER. Joint training is best read as the trade-off point that keeps broad Bengali coverage while recovering most of the code-switch behaviour that Bengali-only training loses — not as a system that dominates on any single metric. The contribution of this paper is the benchmark; the joint-data system is one of three ablations it makes measurable.

Two-sided paired bootstrap tests quantify these condition-specific differences, and human-reference evaluation identifies MMS-1B as best on the current 100-clip subset. Together, these results make the benchmark useful for studying reference bias, dialect variation, English-token retention, and decoder repetition without overstating a single best model. The next priorities are a larger adjudicated human test set, rights-reviewed release governance, targeted collection for underrepresented dialects, and decoding methods that reduce repetition without score-time deletion. Additional priorities are evaluation on Ben-10 and Common Voice, comparison with Bengali-capable SeamlessM4T, targeted collection of code-

switched and underrepresented-dialect speech, and consent-based demographic auditing. These are stated as future work rather than evidence supplied by the current experiments.

DATA AND CODE AVAILABILITY

The dataset landing page and code are available at https://huggingface.co/datasets/niloycste68/Bangali_local_dialect_ASR_HF_Dataset and <https://github.com/niloycste/ASR-Bengali-Local-Dialect->. Audio is under gated access: the release register (ETHICS_AND_DATA_GOVERNANCE.md in the code repository) records three blocking determinations — institutional ethics confirmation, redistribution review, and per-source allow/remove decisions — none of which has yet been made, and hosted audio is not publicly downloadable until they are. Describing this procedure is not a claim to have completed it; the register states its status as of the submission date. The repository documents the multi-stage commands, augmentation step, model identifiers, hyperparameters, split settings, fixed statistical seed, dependency lock file, and checkpoint manifest. Released prediction files reproduce the reported metrics, confidence intervals, and two-sided p -values without retraining.

Because a fixed seed alone does not pin a split — the assignment also depends on the order in which the clip pool is enumerated — we do not ask readers to re-derive the split. The construction script now sorts sources before the seeded, dialect-stratified assignment so that the procedure is deterministic, and the split itself is archived as clip-ID manifests with SHA-256 checksums (`dataset_pipeline/08_export_split_manifests.py`). The test manifest on which every reported number depends is archived in full: 4,088 clips, 176 code-switched, 13.2 hours, matching Table IV. Readers can therefore verify clip membership directly rather than rebuilding the corpus. The train and development manifests must be regenerated from the released dataset and are not byte-guaranteed against the original construction snapshot, which predates the ordering fix; a regenerated release should be versioned rather than assumed identical. End-to-end checkpoint reuse additionally requires depositing the large trained weights at the archived locations listed in the manifest. Audio redistribution is governed by the rights-review procedure described in Section III; where redistribution is not authorized, the release is limited to permissible metadata, source pointers, and processing code.

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